Yonghoon Jeong

Lecturer, Department of Cyber Science
Researcher, Institute of Naval Research
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Research Interests

Wireless Communication Systems

Physical-Layer Wireless Sensing and Communications

Low-Power Wide-Area Networks (LPWANs)

Underwater Acoustic Communications and Ocean IoT

Education

M.S. in Electrical and Computer Engineering 2024 – 2026

- Seoul National University, Seoul, South Korea
- Advisor: Prof. Saewoong Bahk

B.S. in Electronics and Communications Engineering 2018 – 2023

- Kwangwoon University, Seoul, South Korea

Professional Experience

Republic of Korea Naval Academy (ROKNA), Changwon, South Korea 2026 – Present

- Lecturer, Department of Cyber Science
- Researcher, Institute of Naval Research

Seoul National University (SNU), Seoul, South Korea 2024 – 2026

- Researcher, Network Lab (NETLAB), Institute of New Media and Communications (INMC)

Electronics and Telecommunications Research Institute (ETRI), Seongnam, South Korea 2023

- Researcher, AI-SoC Infrastructure Research Section

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea 2022

- Research Intern, Unmanned Systems Research Group (USRG), School of Electrical Engineering (EE)
- Advisor: Prof. David Hyunchul Shim

Publications

Submitted and In-Preparation Manuscripts

- **Yonghoon Jeong**, "On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference," submitted, 2026.
- **Yonghoon Jeong**, "Receiver-Aware Chirp Waveform Optimization for Mobile Underwater LoRa Communications," in preparation, 2026.

International Journal Papers

- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, "On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect," *IEEE Communications Letters*, vol. 30, pp. 2238–2242, Jun. 2026. (SCIE, IF: 4.4) Link
- Dongrack Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, "D^2-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network," *IEEE Internet of Things Journal*, vol. 12, pp. 45325–45340, Nov. 2025. (SCIE, IF: 8.9) Link
- **Yonghoon Jeong** and Youngseok Choi, "Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models," *Biomedical Engineering Letters*, vol. 15, pp. 575–585, Apr. 2025. (SCIE, IF: 3.4)

Link

International Conference Papers

- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, "A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration," *IEEE ICNP 2025*, Seoul, Korea, Oct. 19–22, 2025. Link

Domestic Conference Papers

- **Yonghoon Jeong** and Saewoong Bahk, "A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation," *The Fall Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, Nov. 20–22, 2024.
- **Yonghoon Jeong**, "Development of Non-contact Heart Rate Measurement Technology Based on Facial Video," *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, Jun. 23–25, 2021.

Awards & Scholarships

Scholarship Recipient 2024–2025
- Brain Korea 21 (BK21), National Research Foundation of Korea

Scholarship Recipient 2022–2023
- Korea Scholarship Foundation for the Future Leaders (KOSFFL)

Excellent Paper Awards 2021
- Korea Computer Conference 2021

University President's Award 2020
- Seoul Capstone Design, Seoul

Scholarship Recipient 2020
- Samsong Scholarship Foundation

Dean's List 2020
- Highest GPA

Teaching Experience

Creative Engineering Design, Prof. Saewoong Bahk 2024
Electronic Circuits Experiment I, Emeritus Prof. Young-chul Chung 2023
Electronic Circuits Experiment II, Emeritus Prof. Young-chul Chung 2023
Electronic Circuits Experiment I, Prof. Youngseok Choi 2022

Technical Skills

Programming Languages: C, C++, Python, MATLAB
Frameworks & Libraries: ROS, ROS2, TensorFlow, PyTorch
Simulation Tools: MATLAB/Simulink, ns-3
Development Tools: Linux, Git, Docker